

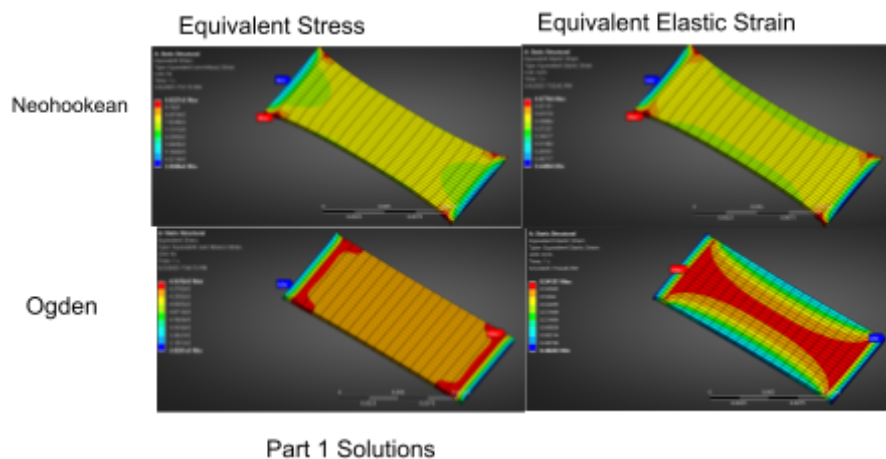
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Mechanical Characterization of Biological Tissues Using Finite Element Models

This report presents a comparison of the mechanical responses of Neo-Hookean and Ogden material models under uniaxial and biaxial tensile tests using finite element analysis. It also explores the impact of different fixation methods (clamp vs. suture) and symmetry boundary conditions in reducing computational complexity.

Part 1: Uniaxial Tensile Test

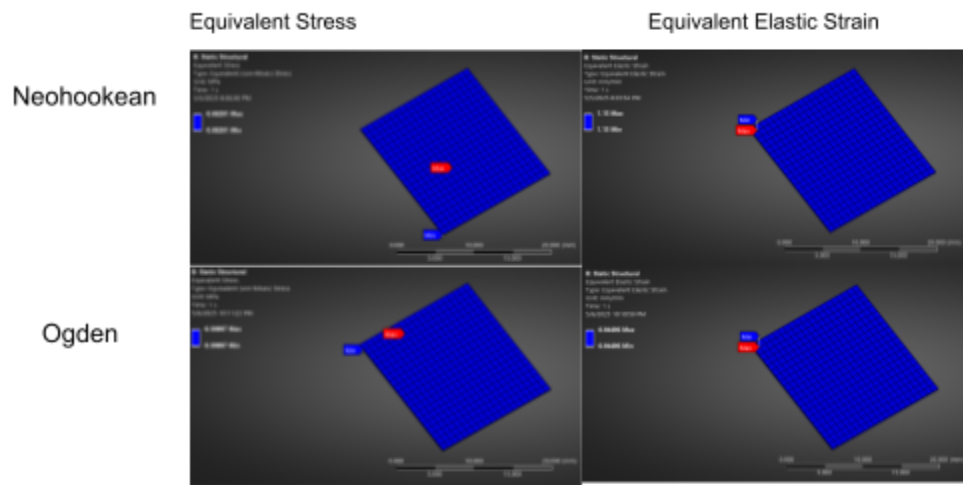
In the uniaxial tensile test, a rectangular biological tissue sample ($10 \times 10 \times 0.5$ mm) was displaced 4 mm outward on opposing sides. The Neo-Hookean model used a shear modulus of 300 kPa, while the Ogden model used a shear modulus of 10 kPa and exponential parameter $\alpha = 8$. Both models assumed near-incompressibility with Poisson's ratio = 0.49. The Neo-Hookean model exhibited a more linear stress-strain response compared to the highly nonlinear Ogden model. Lateral strains (in y and z directions) decreased due to Poisson's effect, while corresponding stresses increased due to volumetric constraints under near-incompressibility. Due to its nonlinearity, the Ogden model required finer time steps to ensure convergence.



Part 2: Biaxial Tensile Test – Clamp vs. Suture Fixation

This test applied 4 mm outward displacements to all four edges. Clamp fixation constrained lateral movement, while suture fixation allowed in-plane deformation. Clamp fixation produced higher stress concentrations, particularly at corners, due to constrained edges. Suture fixation resulted in more evenly distributed stresses and better-represented physiological conditions. Neo-Hookean again displayed linear behavior, while Ogden's

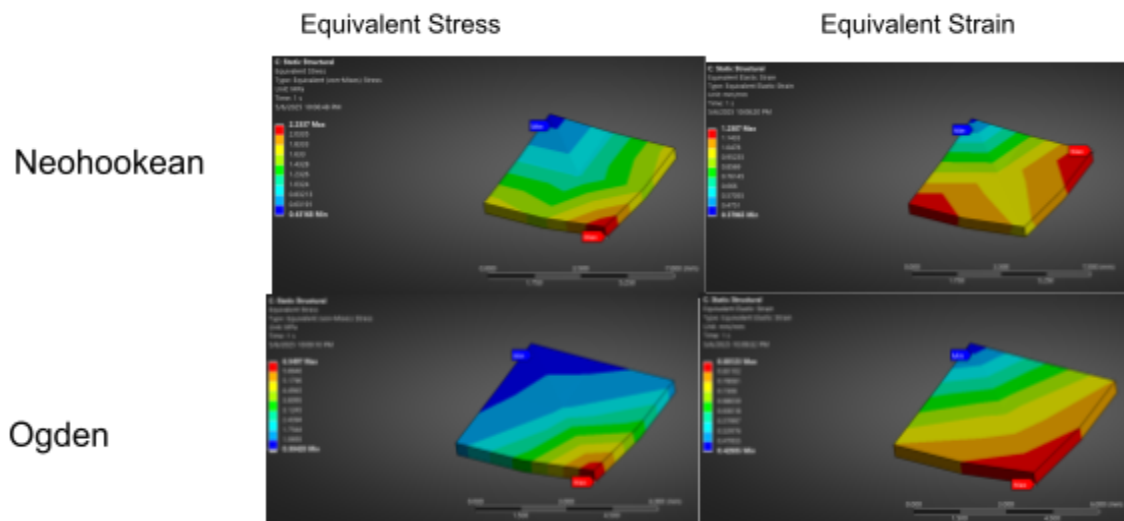
response was highly nonlinear and required stabilization via additional constraints.



Part 2 Figures

Part 3: Symmetry Boundary Conditions

To reduce computation time, a quarter-sample geometry ($5 \times 5 \times 0.5$ mm) was used. Frictionless support was applied to symmetry planes and the bottom face. The same material models and displacement boundary conditions (4 mm outward) were used. Symmetry boundary conditions reduced element count and computation time by $\sim 50\%$ without affecting accuracy. Results closely matched the full model, validating the symmetric approach. Mirroring the post-processed model for visualization remained a challenge in Ansys.



This tutorial demonstrates the tradeoffs between material fidelity and computational efficiency. The Ogden model captures nonlinear soft tissue behavior more accurately but requires careful tuning. Symmetry and suture-based setups can simplify simulations while maintaining physiological relevance. These insights are essential for modeling biological tissues in preclinical device testing or surgical simulation.